

INDR 252 Applied Statistics - Midterm 1 Study Question Solutions

1.

A sample is a subset of the population that can only reflect some characteristics of the population. Due to the sampling variation, samples can have different conclusions and the accuracy can be improved by increasing the sample size or by combining samples. **Answer: a and b**

2.

Bias can be calculated by the formula: $Bias = \bar{X} - True\ value$

Uncertainty of the sample mean can be expressed by standard error which has the formula: $SE = \frac{s}{\sqrt{n}}$

Answer: a

3.

$$E[\hat{\Theta}_1] = E\left[\frac{X+Y}{2}\right] = \frac{1}{2} (E[X] + E[Y]) = \frac{1}{2} 2\mu = \mu \rightarrow \text{unbiased}$$

$$E[\hat{\Theta}_2] = E\left[\frac{X+2Y}{3}\right] = \frac{1}{3} (E[X] + 2E[Y]) = \frac{1}{3} 3\mu = \mu \rightarrow \text{unbiased}$$

$$Var(\hat{\Theta}_1) = Var\left(\frac{X+Y}{2}\right) = \frac{1}{4} Var(X + Y) = \frac{1}{4} (\sigma^2 + \sigma^2) = \frac{\sigma^2}{2} \Rightarrow s_{\hat{\Theta}_1} = \frac{\sigma}{\sqrt{2}} \text{ (X and Y ind.)}$$

$$Var(\hat{\Theta}_2) = Var\left(\frac{X+2Y}{3}\right) = \frac{1}{9} Var(X + 2Y) = \frac{1}{9} (\sigma^2 + 4\sigma^2) = \frac{5\sigma^2}{9} \Rightarrow s_{\hat{\Theta}_2} = \frac{\sqrt{5}\sigma}{3} \text{ (X and Y ind.)}$$

$$MSE(\hat{\Theta}_1) = Bias^2 + s_{\hat{\Theta}_1}^2 = 0 + \frac{\sigma^2}{2} = \frac{\sigma^2}{2}$$

$$MSE(\hat{\Theta}_2) = Bias^2 + s_{\hat{\Theta}_2}^2 = 0 + \frac{5\sigma^2}{9} = \frac{5\sigma^2}{9}$$

Answer: c, d, and g

4.

$$P(X_i = x) = (1 - p)^{x-1} p$$

$$L(p) = P(X_1=2)P(X_2=1)P(X_3=3)P(X_4=2)P(X_5=5)P(X_6=6)P(X_7=1)$$

$$L(p) = (1 - p) p \cdot (1 - p)^0 p \cdot (1 - p)^2 p \cdot (1 - p) p \cdot (1 - p)^4 p \cdot (1 - p)^5 p \cdot p = (1 - p)^{13} p^7$$

Answer: d

5.

$$\ln(L(p)) = \ln((1 - p)^{13}) + \ln(p^7) = 13 \cdot \ln(1 - p) + 7 \cdot \ln(p)$$

$$\frac{d\ln(L(p))}{dp} = -\frac{13}{1-p} + \frac{7}{p}$$

$$\frac{d^2 \ln(L(p))}{dp^2} = -\frac{13}{(1-p)^2} - \frac{7}{p^2} < 0$$

The second derivative is always less than zero, so the first derivative will give the global maximum:

$$\frac{d \ln(L(p))}{dp} = -\frac{13}{1-p} + \frac{7}{p} = 0 \Rightarrow \widehat{p}_{mle} = 7/20$$

Answer: d

6.

The time series of a random sample is expected to have no noticeable patterns or trend. Options b) and d) have clear trends which would be very unlikely to see in a truly random sample. Option c) has a noticeable pattern which also is highly unlikely to be observed in an actual random sample. The best candidate is therefore a), with no pattern or trend to be seen.

Answer: a

7.

7.1

This question is a matter of knowing which z_{α} value and uncertainty measure to use. The exact same formula can be found in the lecture notes.

Answer: a

7.2

$$\alpha = 0.1$$

$$z_{0.05} = 1.645 \text{ (approximately)}$$

$$\bar{x} = 177.9$$

$$\sigma = \sqrt{24.52} = 4.95$$

$$n = 10$$

$$(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}) = (175.32, 180.48)$$

Answer: c

7.3

- a) There is a 90% probability that the population mean μ is a value between L and U.

False: This is not a probability. The population mean is something that already exists, and can be measured, we cannot assign probabilities to things that are either true or false already before observing them.

A simple example: If you asked someone to guess your age, and they said that there is an 80% chance that you are 19 years old, that would make no sense, because you already know whether you are 19 or not.

- b) We can be 90% confident that the population mean μ is between L and U.

True: Confidence and probability are different concepts.

- c) If we increase the confidence level for the same sample, the confidence interval will shrink

False: Intuitively, to have a higher confidence level, we must increase the interval length, otherwise we would just use 100% confidence intervals to get the most precise estimates.

Mathematically, z_{α} increases as $(1 - \alpha)$ increases, so the interval lengths also increase.

- d) If we construct 100 different 90% confidence intervals from 100 different random samples of the population, we expect approximately 90 of them to contain the true population mean.

True: This is true and could even be a definition of what confidence intervals tell us.

- e) Increasing the sample size will always result in a narrower confidence interval for the population mean, assuming we keep the confidence level the same.

True: The formula in 7.1 should make this obvious.

- f) If the standard deviation was not given, we still could have estimated the confidence intervals from the given sample by estimating the standard deviation.

False: The given sample size of 10 is too small to use z-distribution, a t-distribution would be required.

8.

Number of batteries that meet the quality standards = 13, $N = 20$

$$p = \frac{13}{20} = 0.65$$

Traditional CIs:

$$\begin{aligned} & \left(\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right) \\ &= \left(0.65 - z_{0.05/2} \sqrt{\frac{0.65(1-0.65)}{20}}, 0.65 + z_{0.05/2} \sqrt{\frac{0.65(1-0.65)}{20}} \right) \\ &= (0.0441, 0.859) \end{aligned}$$

New version:

$$\begin{aligned} \tilde{p} &= \frac{13 + 2}{20 + 4} = 0.625 \\ & \left(\tilde{p} - z_{\alpha/2} \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}, \tilde{p} + z_{\alpha/2} \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}} \right) \\ &= \left(0.625 - z_{0.05/2} \sqrt{\frac{0.625(1-0.625)}{24}}, 0.625 + z_{0.05/2} \sqrt{\frac{0.625(1-0.625)}{24}} \right) \\ &= (0.431, 0.819) \end{aligned}$$

Since the interval is wider, the new confidence interval provides a more reliable result. This means that a larger proportion of the intervals constructed at this width (closer to the confidence level) will contain the population's success probability, which is the proportion of batteries that meet the quality standards in the population.

Answer: a,b,c, and e

9.

a) The sample is a random sample.

True: If the sample is not random, we cannot estimate anything at all, including using Student's t distribution.

b) The sample size is large.

False: One of the useful things about Student's t distribution is that we can construct confidence and prediction intervals in samples where the sample size is small.

- c) The population standard deviation is known.

False: If the population standard deviation is known, it is better to simply use normal distribution. The t distribution is used to compensate for not knowing the population standard deviation by increasing the interval length for the same confidence levels.

- d) The population is normally distributed.

True: This is a requirement that must be kept in mind if one is to use Student's t distribution.

An Intuition: If the population did not have to be normally distributed but we were allowed to use Student's t distribution anyways, think about what would happen if we were to have $n \rightarrow \infty$ samples. The t distribution looks exactly like a normal distribution as degrees of freedom ($n - 1$) approaches infinity. This means that the population needs to be normally distributed in the first place for Student's t distribution to make analytically correct estimates.

Answer: a and d

10.

- a) The presence of an outlier in a small sample indicates that the sample does not represent the population.

False: The population itself might also have outliers at a similar frequency observed in the small sample.

- b) The Student's t distributions tend not to produce small samples containing an outlier.

False: They are more likely to produce outliers than normal distributions. However, here the sample is not coming from Student's t distribution but from a normal distribution.

- c) Normal distributions tend not to produce small samples containing an outlier.

True: It is unlikely that a small sample that is sampled from a normally distributed population will contain an outlier. The point of t distribution is to be a substitute to z distribution if the population standard deviation is not known and if and only if the population is normally distributed. So, a small sample containing an outlier is likely not

from a normally distributed population, hence it does not make sense to use Student's t distribution in this case.

- d) These methods may be used for a small sample containing an outlier provided the outlier is first discarded.

False: Unless we are certain that the outlier is an erroneous data entry or does not represent the population that is being estimated, we cannot discard the outliers just to use easy estimators. Alternative methods should sadly be sought.

Answer: c

11.

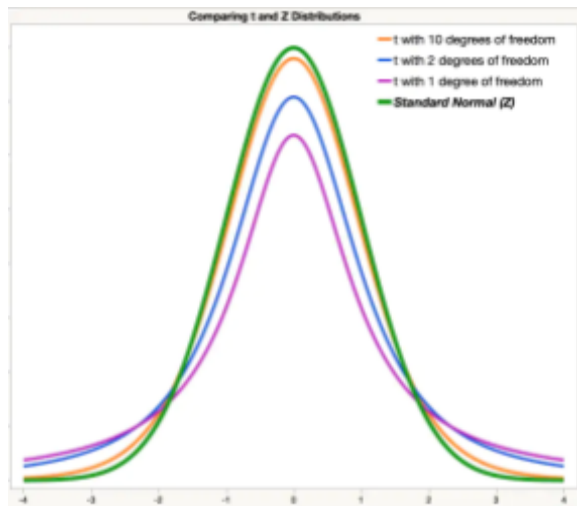
In order for the intervals to contain 0, which corresponds to the fact that the means of two populations are same, the lower bound must have a negative value and the upper bound must have a positive value.

When the t-distribution is used with df of 25, 95% and 99% CIs have a positive upper bound and a negative lower bound, which give the intervals of (-0.839, 0.019) and (-0.991, 0.171), respectively.

$$= \left(\bar{X}_1 - \bar{X}_2 - t_{25, \alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \bar{X}_1 - \bar{X}_2 + t_{25, \alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \right)$$

Answer: a and b

12.



https://www.jmp.com/en_us/statistics-knowledge-portal/t-test/t-distribution.html

Answer: a, b, and d

Answer: a and c

Answer: c

Answer: c

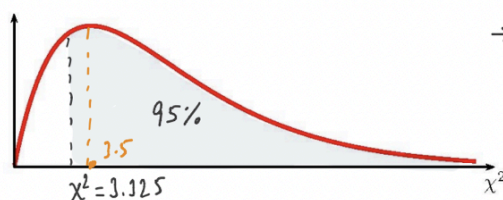
100(1- α)% CI for differences of pairs: $\bar{d} \pm t_{n-1, \alpha/2} \frac{s_d}{\sqrt{n}} = \bar{d} \rightarrow$ differences are exactly 1.

Hence, 2 means are different from each other at all confidence levels.

Answer: b and c

17. The quantity $\frac{(n-1)s^2}{\sigma^2}$ has a chi-square distribution with $(n-1)$ degrees of freedom.

Degrees of freedom (df)	Significance level (α)							
	.99	.975	.95	.9	.1	.05	.025	.01
1	-----	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209



$$\begin{aligned} \rightarrow P(\chi^2 < 3.325) &= 0.05 \\ \Rightarrow P(\chi^2 < 3.5) &> 0.05 \text{ (a)} \\ \text{which also implies} \\ P(\chi^2 > 3.5) &< 0.95 \text{ (d)} \end{aligned}$$

Answer: a and d

18. Since the sample is neither normally distributed nor has a size greater than 40, confidence intervals around the population mean and population variance cannot be found.

Answer: c and d

19. Since the s.d. obtained for both samples are lower than the paired method, it can be concluded that the two samples do not have the same source of variability. Therefore, it is better to use the two-sampled method. Moreover, since the value 0, which indicates the samples are not different, is not included in the 95% CIs, it can be concluded that the two means are different from each other with 95% confidence.

Answer: a and c

20. Since the standard deviation of the differences of pairs is higher than two-sample method's, it cannot be asserted that there exists a high correlation between the two samples. Since no further information is provided, two samples cannot be assumed to be independent.

Having equal mean of differences, since the degrees of freedom of the two-sample method (13) are higher than the paired method's (9), it results in a lower additive term to the mean, which in turn leads to a shorter width. Additionally, the fact that the standard deviations of both samples are lower than the differences of pairs, further contributes to a lower additive term compared to the paired method, also resulting in a shorter width.

Answer: a, c, and d